Rahul Anand Sharma

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Mountain View, CA

Summary

Machine Learning Systems Engineer and Researcher specializing in LLM inference optimization, distributed accelerator scaling (TPUs/GPUs), and hardware-software co-design. Ph.D. in ECE from Carnegie Mellon University with 15+ publications at top systems and security venues (SIGCOMM, NSDI, USENIX Security) and 2 US patents. Track record of leading high-impact technical initiatives across Google Cloud (Gemini inference optimization, TPU v7 Ironwood NPI, disaggregated serving for Sovereign Cloud), AWS (Amazon Q / Generative AI at hyperscale), and Microsoft Research.

Technical Skills

Languages: Python, C++, CUDA, SQL, Bash

ML Frameworks & Compilers: JAX, PyTorch, XLA, Triton, TensorRT / TensorRT-LLM, TensorFlow

LLM Systems & Serving: Disaggregated Serving, vLLM, Model Parallelism (Tensor/Pipeline/Expert), Distributed Training (FSDP, Megatron-LM), KV-Cache Optimization, Speculative Decoding, Quantization (FP8/INT4)

Systems, Cloud & Hardware: Google Cloud TPUs (v4/v5/v7 Ironwood), NVIDIA GPU Clusters, Google Distributed Cloud (GDC), AWS (Glue, EMR, EC2), Apache Spark, Kubernetes, Docker, Ray

Core Domains: LLM Inference Acceleration, Multi-Cube Accelerator Scaling, Compiler-Hardware Co-design, Secure & Isolated AI Infrastructure (Sovereign Cloud / Network Security)

Professional Experience

Google Cloud Aug. 2025 – Present
ML Systems Engineer *Mountain View, CA*

- Driving inference optimization and serving performance for Gemini models across GPU and TPU platforms, focusing on runtime bottleneck analysis, KV-cache efficiency, and end-to-end latency/throughput.
- Led the TPU v7 (Ironwood) NPI microbenchmarking suite; bridged internal cluster infrastructure (Borg) with Cloud XLA runtimes, ported benchmarks from HLO to JAX, and executed multi-cube scaling analyses to model accelerator interconnect limits.
- Delivered collective communication microbenchmarks and public recipes for the TPU v7 public preview, establishing interconnect scaling baselines that advanced multi-billion-dollar hardware evaluations with tier-1 frontier AI customers.
- Spearheaded the intra-node (4:4 split) disaggregated serving architecture for Gemini models on bare-metal nodes in Google Distributed Cloud (GDC), overcoming strict memory isolation and control-plane constraints for Sovereign Cloud.
- Architected automated continuous regression pipelines ensuring Borg-Cloud runtime parity; optimized open-source LLMs (Llama 3, DeepSeek) to land Day-1 3P hardware enablement (Sim-Ship), earning the Google Cloud Tech Impact Award (2026).

AWS, Amazon Jan. 2023 – July 2025
Applied Scientist *Palo Alto, CA*

- Architected and launched a Generative AI-powered auto-debugging system (Amazon Q) for Apache Spark across AWS Glue, Amazon EMR, and MaxDome, utilizing LLMs for automated root-cause analysis to slash Mean Time to Resolution (MTTR) from days to minutes.
- Engineered a high-throughput active testing service for AWS network security, injecting synthetic live traffic to validate Network ACLs at scale, guaranteeing strict compliance and eliminating false-positive blindspots inherent to static simulation models.
- Designed and integrated automated diagnostic tooling directly into AWS Glue and EMR management consoles, delivering real-time failure insights and actionable remediations to thousands of enterprise customers.

Microsoft Research, Redmond May 2020 – July 2021
Research Intern *Redmond, WA*

- Architected low-latency distributed protocols for edge-cloud deep learning inference offloading within the Networking Research Group, minimizing communication overhead and end-to-end latency under bandwidth-constrained networks.

Microsoft Research, India July 2016 – July 2018
Research Fellow *Bangalore, India*

- Co-designed and built FarmBeats, an end-to-end IoT and cloud analytics platform for data-driven agriculture; awarded Microsoft Outstanding Technical Achievement Award and productized into Microsoft Azure (US Patent US20200150640A).
- Pioneered DeepEdge, a novel edge-AI scheduling architecture optimizing deep learning workloads under severe compute and bandwidth constraints, enabling graceful performance degradation (US Patent US10942767B2; published at NSDI).
- Researched neural network learnability with theoretical researchers, analyzing empirical bounds on how data complexity, model depth, and minibatch dynamics govern generalization.
- Mentored 4+ research interns and junior engineers, driving collaborative systems research initiatives that resulted in multiple conference publications.

Education

Carnegie Mellon University — Pittsburgh, PA
Ph.D. in Electrical & Computer Engineering — *Aug. 2018 – Jan. 2023*
- Thesis: *Practical Network Layer Machine Learning for IoT Security*
- Advisors: Prof. Vyas Sekar & Prof. Anthony Rowe | Honors: CIT Dean's Fellow

International Institute of Information Technology (IIIT) — Hyderabad, India
B.Tech and M.S. by Research in Computer Science — *July 2011 – July 2016*
- Honors: Dean's List for Academic and Research Excellence

Patents & Honors

Patents: US10942767B2 (Deep neural network workload scheduling); US20200150640A1 (Sensor fall-curve identification)

Awards & Honors: Google Cloud Tech Impact Award (2026); Microsoft Outstanding Technical Achievement Award (FarmBeats, 2018); CIT Dean's Fellow, CMU (2018); Dean's List, IIIT Hyderabad (2016)

Media & Recognition: Research featured by Bill Gates (*GatesNotes*), Satya Nadella, the front page of *Hacker News*, *The Washington Post*, and *Der Spiegel*.

Selected Publications

15+ peer-reviewed papers at top-tier systems, security, and networking venues with 450+ citations.

USENIX Security 2022 | *Lumos: Identifying and Localizing Diverse Hidden IoT Devices in an Unfamiliar Environment*
– Built a mobile system leveraging 802.11 PHY/MAC features and AR to localize hidden IoT devices with 95% accuracy; covered by Hacker News front page and CyLab.

ACM CoNEXT 2022 | *Lumen: A Framework for Developing and Evaluating ML-Based IoT Network Anomaly Detection*
– Designed an ML evaluation framework across 100+ devices, resolving domain-shift degradation in network traffic anomaly detection.

ACM SIGCOMM 2020 | *Contention-Aware Performance Prediction for Virtualized Network Functions*
– Developed an ML-driven contention-aware predictor for virtualized network functions (VNFs) to guarantee QoS under multi-tenant cloud workloads.

ACM IPSN 2020 | *All that GLITTERs: Low-Power Spoof-Resilient Light Anchors for Augmented Reality*
– Engineered an optical retro-reflective sensing pipeline enabling sub-centimeter AR localization at milliwatt power consumption.

USENIX NSDI 2019 | *DeepEdge: A Network Edge for Deep Learning Workloads*
– Designed a distributed edge-cloud scheduling framework optimizing DNN inference pipelines under tight compute constraints.

Selected Invited Talks

USENIX Security 2022 | "Lumos: Identifying and Localizing Diverse Hidden IoT Devices" (Boston, MA)

CyLab & CONIX Annual Review 2022 | "Lumen: ML-Based IoT Network Anomaly Detection" (CMU, Pittsburgh, PA)

ACM IPSN / CPS-IoT Week 2020 | "All that GLITTERs: Light Anchors for Augmented Reality"

Microsoft Techfest 2018 | "FarmBeats: AI, Edge & IoT for Agriculture" (Redmond, WA)

IEEE WACV 2018 | "Automated Top-View Registration of Broadcast Sports Video" (Lake Tahoe, NV)